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5 DECEMBER 1979
SUPERSEDING
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MILITARY SPECIFICATION

INDICATOR, RATE OF FLOW, FUEL, TYPE EFU-24/A AND TYPE EFU-25/A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers two types of sealed, single remote indicating, synchro, fuel flow rate indicators, range 100 to 2,000 pounds per hour (pph).

1.2 Classification. Indicators shall be furnished in the following type designations and dial markings, as specified (see 6.2):

Type	EFU-24/A	Flourescent dial and pointer markings. (Replaces Type A-16)
Type	EFU-25/A	White dial and pointer markings.

2. APPLICABLE DOCUMENTS

*2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein

SPECIFICATIONS

FEDERAL

PPP-B-636 - Box, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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MIL-P-116	- Preservation - Small Packaging, Methods of
MIL-G-5572	- Gasoline, Aviation, Grades 80/87, 100/130, 115/145
MIL-T-5913	- Transmitter, Rate of Flow, Synchro, Type J-5
MIL-I-7057	- Indicators, Synchro, Aircraft, General Specification For
MIL-L-25142	- Luminescent Material, Flourescent

STANDARDS

FEDERAL

FED-STD-595	- Colors

MILITARY

MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of U S Military Property
MIL-STD-794	- Part and Equipment, Procedures for Packaging and Packing Of
MIL-STD-831	- Test Reports, Preparation Of
MS33585	- Pointer, Dial, Standard Design of Aircraft Instrument
MS33639	- Case, Instrument, Clamp-Mounted Aircraft
MS33678	- Connector, Receptacle, Electric, Integral Mounting

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

*3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (See 4.2 and 6.4).

3.2 General. The requirements of MIL-I-7057 are applicable as requirements of this specification. Additional requirements shall be as specified herein. In the event the requirements of the general specification and this specification conflict, the requirements of this specification shall govern.

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3.3 Design and construction.

3.3.1 Outline dimensions. The outline dimensions of the indicator shall conform to MS33639, for the 2-inch size and the length "L" shall be 3± 0.125 inches.

3.3.2 Case. The case shall be hermetically sealed. A separate nonhermetically sealed cap may be used on the back of the case for the purpose of installing the electrical receptacle.

3.3.3 Indicator dial. The indicator dial shall conform to Figure 1.

3.3.4 Pointer. The pointer shall conform to MS33585-8, except that the length of the pointer shall be such that the tip will overlap the shortest graduation by a distance equal to 1/3 to 2/3 of the length of the shortest graduation.

3.3.4.1 Pointer finish. The shaded portion of the pointer shall be finished with lusterless white, color No. 37875 of FED-STD-595 flourescent-luminescent material in accordance with MIL-L-25142.

3.3.5 Electrical receptacle. The electrical receptacle shall conform to MS33678-14S-2P or MS33678C-14S-2P if a nonhermetically sealed cap is used on the back of the instrument case.

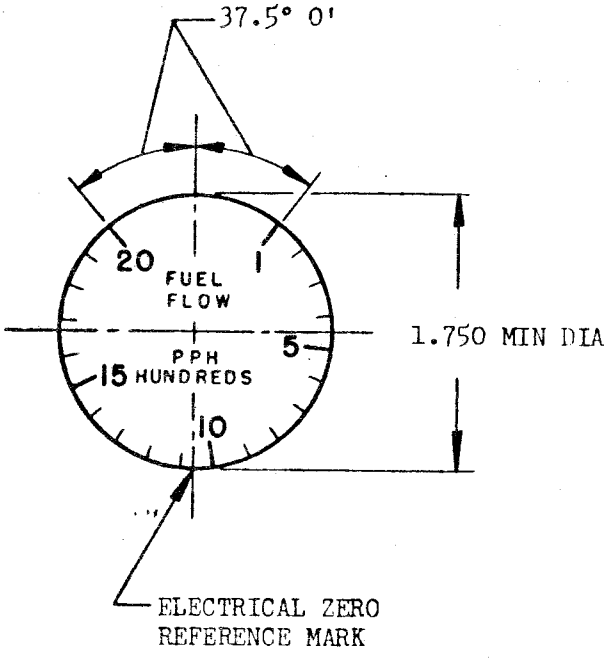
3.4 Calibration. Each indicator shall be calibrated and the dial graduated in accordance with Figure 1. The calibration of fuel flow versus angular degrees shall be linear within ± 0 degrees, 15 minutes of arc.

3.4.1 Each indicator shall be calibrated and adjusted for maximum accuracy at the rate of flow specified in Table I as the "setting point".

TABLE I. Fuel Flow Rate Indicator

Flow rate, pph	Tolerance, pph
100	5.0
200	5.0
400	5.0
600	5.0
800	5.0
1,000	5.0
1,200	5.0
1,400	5.0
1,600	5.0
1,800	5.0
2,000	5.0

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DIMENSIONS IN INCHES. UNLESS OTHERWISE SPECIFIED, TOLERANCES:
ANGLES ±0° 15'

FIGURE 1. Dial

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3.4.2 Unless otherwise specified, the fuel flow rate indicator shall be designed to indicate at a remote station, the rate of fuel flow as transmitted by a type J-5 transmitter in accordance with MIL-T-5913. The indicator shall increase in value when a voltage sequence of ADC is applied at the indicator connector pins.

3.5 Weight. The weight of the complete indicator shall not exceed 0.70 pound.

3.6 Markings.

3.6.1 Dial Markings. The following markings shall be finished in lusterless white, color No. 37875 of FED-STD-595 flourescent-luminescent material, or in accordance with MIL-L-25142. The dimensions of the marking shall be:

	Height or length, (inch + 0.016	Width of line or graduation, (inch) + 0.005
Scale numerals 1, 5, 10, 15 and 20	0.156	0.031
Graduations opposite numerals 1, 5, 10, 15, and 20	.156	.031
100-pound graduations	.094	.020
Lettering "PPH"	.062	.016
Lettering "HUNDREDS"	.062	.016
Lettering "FUEL FLOW"	.094	.020

3.6.1.1 Color. The marking "Type EFU-24/A" or "Type EFU-25/A", as applicable, shall be permanently and legibly marked on the dial. These markings, and all other markings not otherwise specified, shall be finished in lusterless black, color No. 37038 of FED-STD-595. The height of the markings shall be 0.047 ± 0.016 inch.

3.7 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130. The following special characteristics shall be added:

26V, 400 Cycles

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the contractor may utilize his own facilities or any commercial laboratory acceptable to the Government.

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The Government reserves the right to perform any of the inspections set forth in the specifiction where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

*4.2 Classification of Inspection. Qualificaion inspection, quality conformance inspection and test conditions shall be in accordance with MIL-I-7057 and as specified herein.

4.3 Quality conformance inspection. The quality conformance inspections shall consist of the following individual and sampling tests. Performance of the indicator for scale error and friction shall be checked against Tables I and II of this specification.

4.3.1 Individual tests. Each indicator submitted for acceptance under contract shall be subjected to the individual tests specified in MIL-I-7057.

4.3.2 Sampling tests. Sampling tests shall be as specified in accordance with MIL-I-7057. Failure of any sample to meet requirements therein, shall cause rejection of the lot.

*4.3.3 Test Samples shall consist of three indicators representative of production equipment. Samples shall be identified with the manufacturer's part number and such other information as required by the procuring activity.

*4.3.4 Test Report. After the contractor completes the qualification tests he shall prepare a test report in accordance with MIL-STD-831 and furnish three complete copies to the procuring activity.

TABLE II. Fuel Flow Indicator

Test	Tolerance, pph
Position error at 1,050 pph	2
Friction:	
Room temperature	10
High temperature	10
Low temperature	20
Salt Spray	4
Vibration:	
Pointer oscillation	5
Pointer variation	5
Scale error:	
Low temperature	7
High temperature	5
Salt spray	4

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4.4 Inspection and test. Test methods of preservation and packaging shall be accomplished in accordance with Section 4 of MIL-P-116 to insure compliance with Section 5 of this specification.

5. PACKAGING

*5.1 Preservation-Packaging. Preservation-packaging shall be Level A or C as specified.

*5.1.1 Level A

*5.1.1.1 Cleaning. Indicator, Rate of Flow, Fuel, shall be cleaned in accordance with process C-1 of MIL-P-116.

*5.1.1.2 Drying. Indicator, Rate of Flow, Fuel shall be dried in accordance with process D-4 of MIL-P-116.

5.1.1.3 Preservation Application. Not applicable.

5.1.1.4 Unit Packaging. Unless otherwise specified by the procuring activity, each Indicator, Rate of Flow, Fuel, shall be packaged in quantity unit packs of one each in accordance with Method 1C-1 of MIL-P-116. Overbox in PPP-B-636 carton large enough to allow for application of sufficient cushioning material between container and bag, of a type, density, and thickness to insure shock transmission does not exceed peak values in Gs stablished for the item when completed packs are subjected to rough handling drop tests of MIL-P-116.

5.1.2 Level C. Same as Level A except Method III of MIL-P-116 applies in lieu of specified method.

5.2 Packing. Packing shall be Level A, B, or C as specified.

5.2.1 Level A. Indicator, Rate of Flow, Fuel, packaged as specified in 5.1.1 shall be packed in shipping containers conforming to PPP-B-636, weather resistant class, unless otherwise specified by the procuring activity. Insofar as practical, exterior shipping container shall be of uniform shape, size, of minimum tare and cube consistent with the protection required. Table II of MIL-STD-794 applies for Army use.

5.2.2 Level B. Indicator, Rate of Flow, Fuel, packaged as specified in 5.1.1 shall be packed in domestic class exterior shipping container conforming to PPP-B-636. Other requirements as specified in 5.2.1 above shall apply.

5.2.3 Level C. Packing shall be applied which affords adequate protection during domestic shipment from the supply source to the first domestic receiving activity for immediate use. This level

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shall conform to applicable carrier rules and regulations and may be the contractor's commercial practice provided the latter meets the requirements of this level.

5.3 Marking. In addition to any special marking required by the contract, each unit package and shipping container shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The fuel flow indicator, in conjunction with the J-5 transmitter, is intended to indicate the rate of flow of fuel in pph being consumed by an aircraft engine.

6.2 Ordering data. Procurement documents should specify:

(a) Title, number, and date of this specification.
(b) Data requirements
(c) Dial and pointer finish desired (see 3.3.4.1 and 3.6).
(d) Applicable level of packaging and packing (see 5.1 and 5.2).
(e) Type designation (see 1.2).

6.3 Qualification With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in the applicable Qualified Products List whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contacts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is HQ AFLC CASO/LODS, Federal Center, Battle Creek, MI 49016 and information pertaining to qualification of products may be obtained from that activity.

6.4 Reclaimed materials. The use of reclaimed materials shall be encouraged to the maximum extent possible.

6.5 Marginal indicia. The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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MIL-I-5914, Indicator, Rate of Flow, Fuel, Type EFU-24/A and EFU-25/A

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

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